IntelliPath

Project
Proposal

Meet the team

Alex Pulikkottil

Matilda Minasian

Jerry Ttuyan

Project Main Idea

The main idea of this project is to develop a navigation application that integrates real time traffic data and applies machine learning algorithms to learn from historical and live traffic patterns. Unlike traditional navigation systems that only react to current congestion, this app aims to predict traffic conditions and optimize routes proactively.

Problem Formulation

- Input: Historical and live traffic data
- Output: Future road speed/travel time
- Task: Supervised regression

Commuting in traffic sucks…

Los Angeles traffic follows strong recurring patterns such as rush hours, school pickup times, and major events. Most navigation systems react to congestion only after it happens, rather than anticipating predictable traffic behavior.

This project aims to transform commuting into a smarter, less stressful experience by developing a navigation system that learns from both historical and live traffic data. By incorporating machine learning, the system can anticipate congestion and adapt route recommendations proactively.

The core objective is to evaluate whether learning-based routing can outperform traditional static or greedy approaches when future traffic uncertainty is considered—ultimately reducing travel time and improving overall efficiency.

Project Method

Step 1

Step 1 — Traffic Prediction
Use historical + live traffic data to predict future road speeds.

Step 2

Step 2 — Predictive Routing
Use A* algorithm with predicted travel times to compute routes.

Step 3

Step 3 — Learning-Based Optimization (optional)
Apply bandit learning to adapt route selection over time.

Goal

Predictive, learning based route optimization

System Architecture

Traffic Data Sources

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Traffic Prediction Model

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Predicted Road Travel Times

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Road Network Graph

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A* Routing Algorithm

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Best Route Recommendation

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(Optional) Learning-based Route Adaptation

Project Dataset

Live traffic Data

TomTom Traffic APIs

INRIX Traffic Data

Historical Data

Los Angeles GeoHub

Caltrans PeMS

Open Map Data

OpenStreetMap
METR-LA Dataset

METR-LA data

- Traffic sensor data from Los Angeles highways
- Widely used benchmark for traffic forecasting
- Used to train traffic prediction models

Feature Engineering

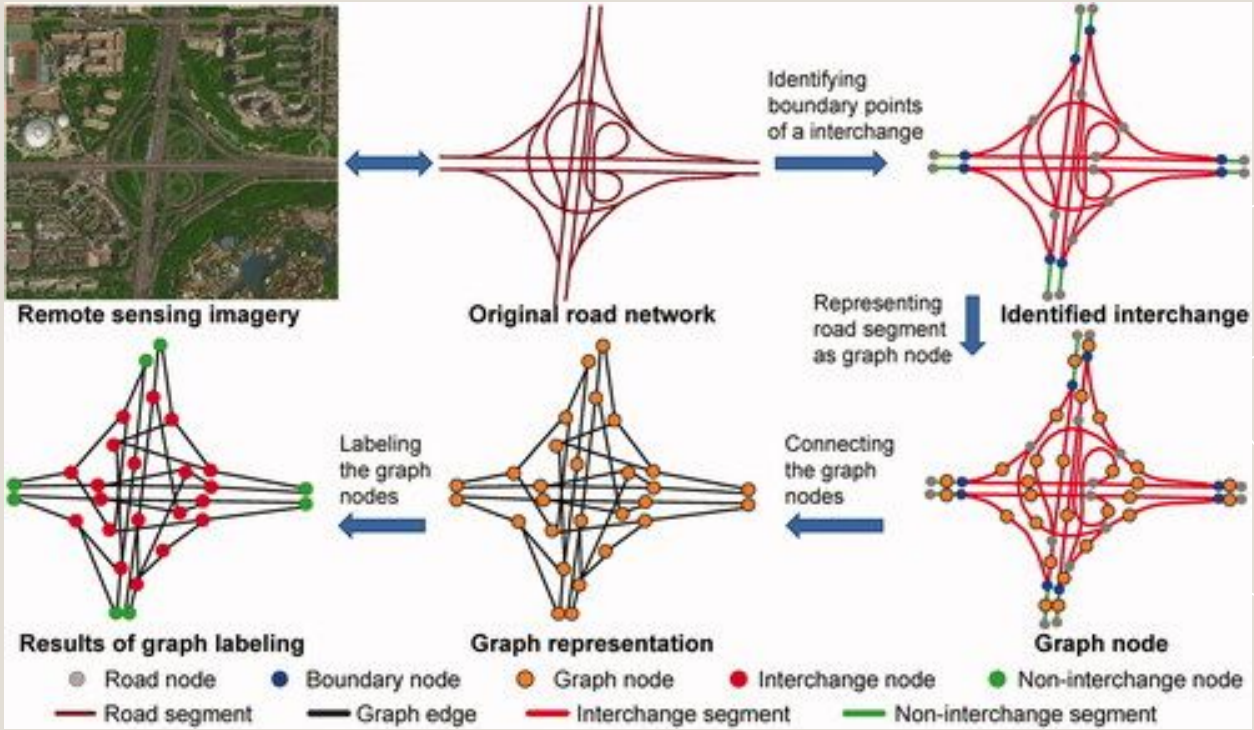
- Time of day (rush hour patterns)
- Day of week (weekday vs weekend)
- Historical traffic speeds (recent time windows)
- Neighboring sensor data (spatial correlation)

Model Choice

- Model:
  - Linear Regression

- Why:
  - Simple and interpretable
  - Fast to train and evaluate
  - Strong baseline for comparison

Graph Representation



Routing Algorithm

Algorithm: A* Search

Why A*?

- Efficient shortest-path algorithm
- Uses heuristics to speed up search
- Widely used in navigation systems

Programming Language & Tools

- **Python**
- Libraries:
  - NumPy / Pandas (data processing)
  - NetworkX or similar (graph routing)
  - Scikit-learn (statistical models)
  - PyTorch or simple bandit algorithms (learning component)

Experiment Plan

1. Build a baseline routing system using current traffic only.
2. Add statistical traffic forecasting.
3. Implement learning-based route selection.
4. Compare performance across different traffic scenarios and time periods.

Baseline Comparison

- Shortest path (with no traffic)
- Current traffic only
- Our method (ML + A*)

Expected Outcome of the Project

The expected outcomes include:

- A routing system that avoids roads likely to become congested in the near future
- Improved average travel time compared to baseline routing
- More reliable routes with lower worst-case delays
- Demonstration that learning-based route selection adapts better to traffic patterns than static approaches

Evaluation Strategy

- Train/ validation/ test split
- Evaluate on unseen data
- Metrics
  - Avg travel time
  - Worst-case

Criteria for Evaluation

The system will be evaluated using:

- **Average travel time**
- **Worst-case (90th percentile) travel time**
- **Travel time regret** compared to the best route in hindsight
- Comparison against:
  - Shortest-path routing
  - Current-traffic-only routing

Related Work / Background Materials

If the project scope is adjusted or simplified, the following references provide foundational support:

1. **Time-Dependent Shortest Path Algorithms**
   – Cooke & Halsey, “The Shortest Route Through a Network with Time-Dependent Internodal Transit Times”

2. **Traffic Prediction Using Historical Data**
   – Vlahogianni et al., “Short-term traffic forecasting: Where we are and where we’re going”

3. **Learning-Based Routing / Bandits**
   – Agrawal & Goyal, “Thompson Sampling for Contextual Bandits”

These works motivate the use of predictive modeling and learning-based decision making in routing problems.

Summary

This project explores how statistical traffic prediction combined with reinforcement learning can produce smarter route recommendations that anticipate congestion rather than react to it. By focusing on Los Angeles traffic patterns and individual ETA optimization, the project demonstrates a practical application of machine learning to real-world transportation systems.